Haitao Chen

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Summary

Ph.D. candidate in Biomedical Engineering at Duke University's Computational Optics Lab, working at the intersection of optics and machine learning. Builds multi-camera microscopes that capture gigapixel-scale images and video in a single snapshot, designs the optics and the reconstruction algorithms together, and pairs the hardware with physics-informed learning for 3D/4D reconstruction and high-throughput imaging of biological dynamics.

Education & Teaching

Duke University, Durham, NC, USA *Aug 2024 – Present*
Ph.D. in Biomedical Engineering (expected Jun 2029) | Advisor: Dr. Roarke Horstmeyer

Huazhong University of Science and Technology (HUST), Wuhan, China *Sep 2020 – Jun 2024*
B.Eng. in Biomedical Engineering | Advisor: Dr. Yong Deng

Teaching Assistant, Duke University, graduate course *Machine Learning and Imaging* *Jan 2026 – May 2026*

Research Experience

Computational Optics Lab, Duke University, Durham, NC, USA *Aug 2024 – Present*
Research Assistant | Advisor: Dr. Roarke Horstmeyer

- Multi-camera array microscopy: assembled, focused, and calibrated 150 cameras across four synchronized arrays for gigapixel-scale imaging at video rates (up to 709 MP per frame and 4.7 gigapixels/s).
- Curvature-adaptive gigapixel microscopy (PANORAMA): per-camera refocusing for 0.84 µm resolution across a 16.3 × 18.8 mm field in a single snapshot (*Optics Letters* 2025, Editors' Pick).
- Optics-algorithm co-design: Fourier light-field architectures, phase masks co-designed with ray-tracing and PSF models, differentiable optical models, and neural-field 3D/4D reconstruction in JAX.
- High-throughput 3D behavioral phenotyping of zebrafish larvae with dual-view HRNet tracking (*Communications Biology* 2026); GPU ISP pipelines for gigapixel RAW streams.
- Next-generation 24-camera array system for Ramona Optics' commercial microscopes (industry collaboration).

Light, Sound, and AI Lab, Rice University, Houston, TX, USA *Aug 2023 – Apr 2024*
Visiting Researcher (full-time, during the final B.Eng. year) | Host: Dr. Lei Li

- Kilohertz computational photoacoustic tomography with physics-informed, label-free 3D reconstruction (B.Eng. thesis).

Britton Chance Center for Biomedical Photonics, HUST, Wuhan, China *May 2021 – Jun 2024*
Research Assistant | Advisor: Dr. Yong Deng

- Physics-informed deep learning for inverse problems in biomedical optics: fluorescence diffuse optical tomography (*Optics Express* 2025) and tissue optical-property estimation (*Biomedical Optics Express* 2023).

Publications

Journal Articles (*equal contribution)

1. **H. Chen***, K. Li*, L. Kreiss, P. Reamey, et al. "High-throughput multi-camera array microscope platform for automated 3D behavioral analysis of swimming zebrafish larvae," *Communications Biology* 9, 141 (2026). [Paper] [Code]
2. X. Yang*, **H. Chen***, L. Kreiss, C. B. Cook, et al. "Curvature-adaptive gigapixel microscopy at submicron resolution and centimeter scale," *Optics Letters* 50, 5977–5980 (2025). Editors' Pick. [Paper]
3. **H. Chen***, K. Liu*, Y. Jiang, Y. Dou, and Y. Deng. "Multierror learning enhanced fluorescence diffusion tomography in reflection geometry," *Optics Express* 33, 22062–22077 (2025). [Paper]
4. **H. Chen**, K. Liu, Y. Jiang, Y. Liu, and Y. Deng. "Real-time and accurate estimation ex vivo of four basic optical properties from thin tissue based on a cascade forward neural network," *Biomedical Optics Express* 14, 1818–1832 (2023). [Paper]
5. K. Li, X. Yang, J. Bagwell, L. Kreiss, **H. Chen**, et al. "High-throughput heart rate sensing in freely moving zebrafish using a multi-well plate-based fluorescence microscope," *Optics Letters* 51, 5269–5272 (2026). [Paper]
6. K. Liu, Y. Jiang, W. Li, **H. Chen**, Q. Luo, and Y. Deng. "High-fidelity mesoscopic fluorescence molecular tomography based on SSB-Net," *Optics Letters* 48, 199–202 (2023). [Paper]
7. J. R. Alonso, B. Javidi, A. Anand, M. Arocena, Y. Awatsuji, S. Carney, R. Castañeda, **H. Chen**, et al. "Roadmap on quantitative phase imaging in biomedical applications," submitted to *Journal of Physics: Photonics*.

Book Chapters (*equal contribution)

1. J. Lerner*, **H. Chen***, L. Kreiss, M. Hoffmann, and R. Horstmeyer. "Wavefront engineering and adaptive optics for deep brain imaging," submitted to *Illuminating the Depths: Advanced Technologies in Deep Brain Imaging Using Light and Sound*.
2. **H. Chen**, J. F. Lerner, A. Chaware, and R. Horstmeyer. "High resolution, wide field-of-view microscopy," in preparation for *Encyclopedia of Modern Optics*.

Conference Papers, Presentations, and Demonstrations

1. **H. Chen**, K. Li, L. Kreiss, et al. "High-throughput automated 3D behavioral profiling of freely swimming zebrafish larvae using a multi-camera array microscope," SPIE Photonics West (BiOS), Proc. SPIE PC13862, PC1386203 (2026). Oral, BaySpec Best Presentation Award. [Video]
2. **H. Chen**, X. Yang, L. Kreiss, G. Kuczewski, M. O. Bohlen, and R. Horstmeyer. "Centimeter-scale gigapixel microscopy with curvature-adaptive focusing and submicron resolution," SPIE Photonics West (BiOS), Proc. SPIE 13859, 1385908 (2026). Oral. [Video]
3. S. J. Tang, C. B. Cook, J. Foust, **H. Chen**, H. Shi, M. Harfouche, C. Toth, A. Kuo, and R. W. Horstmeyer. "Real-time 3D reconstruction and display of synchronized Fourier lightfield video for novel microsurgical applications," Proc. SPIE 13838, 1383806 (2026).
4. C. Cook, K. Kim, J. Lerner, **H. Chen**, K. Zhou, et al. "Compact, synchronized camera arrays for new microscopy applications and beyond," IEEE International Conference on Computational Photography (ICCP 2025). Live demonstration.
5. **H. Chen**, et al. "High-throughput multi-camera array microscope platform for automated 3D behavioral analysis of freely swimming zebrafish larvae," Strategic Conference of Zebrafish Investigators (2025). Poster.
6. **H. Chen**, et al. "Full optical properties estimation method for thin ex vivo tissues," China Biomedical Engineering Conference & Medical Innovation Summit (2023). Oral.

Awards and Honors

- BaySpec Best Presentation Award, SPIE Photonics West (BiOS) *2026*
- Best Bachelor's Thesis Award, Huazhong University of Science and Technology *2024*
- Outstanding Graduate, Huazhong University of Science and Technology *2024*
- Innovation and Technology Scholarship, Huazhong University of Science and Technology *2023*
- National Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM) *2022*
- National Undergraduate Innovation and Entrepreneurship Training Program *2022*

Professional Service

Peer Reviewer *2025 – Present*

Independent reviewer for journals including *Photonics Research*, *ACS Photonics*, *Optics Letters*, *IEEE Transactions on Industrial Informatics*, *Biomedical Optics Express*, and *Journal of Biomedical Optics*, and for the IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI 2025)

Media Coverage

The curvature-adaptive gigapixel microscope (*Optics Letters* 2025) was covered by Optica, Phys.org, EurekAlert!, Photonics Spectra, Optics.org, Nanowerk, LabMedica, BioTechniques, Bioengineer, and F4 News.

Skills

Optics & Imaging: Optical system design and prototyping, multi-camera arrays, light-field and computational imaging, optics-algorithm co-design; Zemax, SolidWorks
Machine Learning & Vision: Physics-informed and self-supervised learning, neural fields, 3D/4D reconstruction, pose tracking
Programming: Python (PyTorch, JAX), MATLAB, C/C++, Git